

PROJECT TOPIC: ECOLOGICAL STUDY OF A FRESHWATER LAKE ECOSYSTEM

Topic Description: Field Study of an Urban Freshwater Lake Ecosystem: Biodiversity, Water Quality Parameters, Eutrophication Threats, and Conservation Strategies

Freshwater aquatic ecosystems—encompassing ponds, natural wetlands, reservoirs, and urban lakes—are among the most biologically productive yet intensely threatened habitats on Earth. In semi-arid and rapidly urbanizing landscapes across Maharashtra, natural and historical man-made lakes function as crucial hydrological sponges that recharge ground aquifers, mitigate urban flash floods, regulate local microclimates, and sustain complex aquatic food chains.

This project presents a structured ecological field investigation of an **Urban Freshwater Lake Ecosystem** (surveying a representative municipal water body such as Pashan Lake in Pune or Powai Lake in Mumbai). The study investigates the macroscopic and microscopic biodiversity supported across different ecological zones of the lake, ranging from littoral emergent marshes to the open limnetic water column.

Furthermore, the project documents severe anthropogenic pressures threatening urban lakes: untreated domestic sewage discharge, cultural immersion debris, municipal encroachment, plastic pollution, and uncontrolled eutrophication driven by invasive water hyacinth (*Eichhornia crassipes*). Practical community restoration measures are synthesized to ensure sustainable wetland preservation.

2. Importance of the topic (विषयाचे महत्त्व)

Documenting the ecological dynamics of a local freshwater lake provides essential educational and civic insights into urban hydrology:

Key Ecological Services of Freshwater Lakes :-

- **Urban Hydrological Buffering:** Lakes act as natural detention basins during intense monsoon cloudbursts, absorbing millions of liters of stormwater runoff and preventing catastrophic urban street flooding.
- **Groundwater Recharge & Aquifer Maintenance:** Continuous percolation through permeable lake beds maintains hydrostatic head pressure in surrounding open wells and borewells within a 3-kilometer radius.
- **Refuge for Aquatic Biodiversity & Migratory Avifauna:** Lake ecosystems support a rich trophic pyramid: phytoplankton, zooplankton, aquatic insects, mollusks, freshwater fish, amphibians, and resident as well as trans-continental wintering migratory waterbirds.
- **Urban Microclimate Regulation:** Evaporative cooling from large lake surfaces moderates surrounding urban heat island (UHI) temperatures by 1.5°C to 2.5°C during hot summer months.

Lakes are living ecological indicators reflecting the environmental health of an entire urban catchment.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

This field project was designed to fulfill structured biological, limnological, and conservation objectives:

Structured Project Objectives :-

- • **Zonational Biodiversity Mapping:** To observe, document, and identify flora and fauna inhabiting the littoral (shallow shoreline), limnetic (open water), and profundal/benthic (lakebed) zones.
- • **Trophic Food Web Construction:** To construct an authentic freshwater aquatic food chain and food web linking autotrophic aquatic plants to top avian predators.
- • **Physico-Chemical Threat Assessment:** To record observational water quality indicators, including water clarity (Secchi depth), odor, dissolved organic foam, and surface oil sheens.
- • **Eutrophication & Invasive Species Audit:** To estimate the surface area coverage of invasive Water Hyacinth (*Eichhornia crassipes*) and determine its suffocating effect on dissolved oxygen levels.
- • **Community Restoration Strategy:** To formulate practical bioremediation, citizen cleanup, and municipal buffer-zone conservation recommendations.

The scope encompasses direct morning field surveys, photographic logs, and water sample observations.

4. Project Work Methodology (प्रकल्प कार्यपद्धति)

A standardized observational ecological and limnological survey protocol was deployed to record lake parameters:

Field Methodology Phases Followed :-

- **Phase 1 – Preliminary Reconnaissance & Site Selection:** Selected an accessible local lake and designated three distinct 50-meter shoreline survey transects: inlet storm drain, open littoral zone, and undisturbed weed bed.
- **Phase 2 – Direct Biological Observation:** Conducted early-morning field visits (6:30 AM to 9:30 AM) with 8x42 binoculars and field spotting guides to document bird, amphibian, and odonate (dragonfly) species.
- **Phase 3 – Physical Water Quality Testing:** Collected surface water samples in clear glass jars to observe turbidity, color, suspended solids, and algal blooms; measured water temperature across sampling points.
- **Phase 4 – Anthropogenic Threat Mapping:** Identified point-source sewage discharge pipes, plastic litter accumulations, construction debris dumping, and weed infestation density.
- **Phase 5 – Synthesis & Syllabus Alignment:** Organized observational findings into standard tables and correlated results with Chapter 7 EVS competencies.

All recorded bird and insect species were cross-verified using BNHS field guide manuals.

5. Observations (निरीक्षणे)

Systematic field observations recorded across three sampling stations revealed rich biological diversity co-existing with severe ecological stress:

5.1 Biodiversity & Ecological Status Matrix of Surveyed Lake

Ecological Category	Observed Species / Indicators	Ecological Role & Condition Observed
Aquatic Flora	Lotus (<i>Nelumbo nucifera</i>), Hydrilla, Typha (Cattails), Water Hyacinth (invasive)	Primary producers; emergent Typha provides bird nesting cover, but Water Hyacinth blankets 45% of surface, blocking sunlight.
Invertebrates & Insects	Daphnia (water fleas), Water Strider, Green Darner Dragonfly, Apple Snail (<i>Pila globosa</i>)	Secondary consumers and detritivores; healthy dragonfly nymph presence indicates moderate shoreline benthic health.
Ichthyofauna (Fishes)	Rohu, Catla, Common Carp, invasive Tilapia (<i>Oreochromis niloticus</i>)	Tilapia dominance over indigenous carps indicates high tolerance for organically polluted, low-oxygen waters.
Avifauna (Waterbirds)	Indian Spot-billed Duck, Little Cormorant, White-throated Kingfisher, Purple Swamphen, Pond Heron	Apex carnivores; 32 resident waterbirds counted feeding on littoral fish, frogs, and submerged aquatic vegetation.
Anthropogenic Threats	Raw municipal inlet storm sewer, single-use plastic bottles, immersion silt, weed choked outlet	Visible gray foam and anaerobic hydrogen sulfide odor near stormwater inflow pipe; severe eutrophication detected.

5.2 Constructed Lake Food Chain

Solar Energy → Phytoplankton / Algae → Zooplankton (Daphnia) → Small Fish (Puntius) → Little Cormorant / Kingfisher.

6. Analysis of data (माहितीचे विश्लेषण)

Detailed analysis of the ecological data illustrates the rapid onset of cultural eutrophication in urban freshwater lakes:

Core Analytical Inferences :-

- **Mechanism of Cultural Eutrophication:** Inflow of untreated domestic sewage introduces massive amounts of nitrogen and phosphorus from detergents and sewage. This nutrient overload fuels exponential blooms of invasive Water Hyacinth (*Eichhornia crassipes*), which doubles its biomass every 10 days.
- **Deoxygenation & Benthic Suffocation:** When thick hyacinth mats cover the water surface, atmospheric gas exchange is completely blocked. When the weed dies, decomposing aerobic bacteria consume all dissolved oxygen (DO), driving levels below 2 mg/L and suffocating native carps and benthic snails.
- **Loss of Native Biodiversity to Hardy Exotics:** Invasive exotic Tilapia dominate the lake because they can breathe atmospheric oxygen at the surface and tolerate organic filth, displacing vulnerable indigenous fish like Rohu and Mrigal.
- **Sedimentation & Loss of Flood Buffer Capacity:** Decades of untreated storm silt and post-festival clay immersion deposits have reduced the average lake depth from 4.5 meters to under 1.8 meters, severely compromising flood retention capacity.

The analysis confirms that urban lakes cannot survive as self-cleaning bodies without strict sewage interception.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Urban freshwater lakes are irreplaceable natural assets that provide vital biodiversity refuges, microclimate cooling, and stormwater flood moderation. However, without immediate sewage interception and community-driven weed removal, our local lakes face imminent biological death and complete swamp succession.

Action Checklist for Lake Conservation :-

- ❑ Demand 100% municipal sewage interception and biological treatment before runoff enters urban lakes.
- ❑ Mandate eco-friendly clay idols and artificial immersion tanks to prevent toxic heavy metal lake contamination.
- ❑ Organize community volunteer weekend cleanup drives to physically remove floating plastic debris.
- ❑ Plant native riparian vegetation (Vetiver grass, Bamboo, Arjun) along shorelines to filter incoming runoff.
- ❑ Establish citizen 'Lake Vigilance Committees' to monitor and report illegal construction encroachments.

8. Bibliography / References (संदर्भ सूची)

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